FAQs

1. What is binary search?

Ans: Binary search is a searching algorithm that repeatedly divides a sorted array into two halves to find the required element. It compares the key with the middle element and searches only the required half.

2. What is the condition for using binary search?

Ans: The array must be sorted in ascending or descending order. In our program, the array should be sorted in ascending order for binary search to work correctly.

3. What is the difference between iterative and recursive binary search?

Ans: Iterative binary search uses loops such as `while` or `for` to repeat the searching process. Recursive binary search uses function calls, where the function calls itself to search in the next half.

4. What happens if the element is not in the array?

Ans: If the element is not found, the function returns `-1`. This happens when the search range becomes empty, that is, when `low > high`.

5. What is the time complexity of binary search?

Ans: The worst-case time complexity of binary search is O(log n) because the search area is reduced to half after every comparison. Its best-case time complexity is O(1) when the key is found at the middle position in the first comparison.

6. Can binary search be used on an unsorted array?

Ans: No, binary search requires the array to be sorted. If the array is unsorted, it may eliminate the wrong half and give an incorrect result.